Final Exam Summary: The Financial Cycle & Yield Curve

> Sources: PIMCO Yield Curve Basics | FRBSF Economic Letters (2003, 2013, 2022) | Project Guidelines

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1. The Yield Curve — Fundamentals

What Is the Yield Curve?

A line graph plotting **bond yields vs. time to maturity** for bonds of the same asset class and credit quality (typically U.S. Treasuries). It captures the **term structure of interest rates**.

- **X-axis**: Maturity (3 months → 30 years)
- **Y-axis**: Yield to maturity (%)
- Most widely used: **U.S. Treasury yield curve** (no credit risk; covers all maturities)

Two Key Yield Concepts

Concept	Definition
Current Yield	Annual coupon ÷ Purchase price
Yield to Maturity (YTM)	Total return if held to maturity (includes price appreciation/depreciation + interest-on-interest)

> YTM is more meaningful for comparing bonds of different characteristics.

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2. Yield Curve Shapes & Economic Signals

Shape	Characteristic	Economic Signal
Normal (Upward Slope)	Long-term yields > Short-term yields	Healthy economic growth expected
Steep	Large gap: long vs. short yields	Strong economic recovery/upturn ahead

Shape	Characteristic	Economic Signal
Flat	Short Long yields	Economic slowdown; transitional phase
Inverted (Downward Slope)	Short-term yields > Long-term yields	**Recession warning** (typically 12–18 months ahead)

Historical Examples

- **Steep curve** (April 1992): Signal of U.S. recovery from 1990–91 recession
- **Flat curve** (December 1989): Preceded 1990–91 recession
- **Inverted curve** (March 2000): Preceded 2001 recession

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3. What Determines Yield Curve Shape — Three Theories

A. Pure Expectations Theory

- Slope reflects **only expectations of future short-term rates**
- Upward slope = markets expect rates to rise

B. Liquidity Preference Theory

- Long-term yields include a **term (liquidity) premium** on top of rate expectations
- Investors demand extra yield for locking up capital longer → normally upward slope

C. Preferred Habitat Theory

- Investors have **preferred maturity "habitats"**; require premiums to move outside them
- Short-term investors dominate → long-term yields tend to be higher

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4. What Makes the Yield Curve Move? (FRBSF, 2003)

Three Statistical Factors (explain >99% of yield movements)

Factor	Effect on Yield Curve
Level	Parallel shift — all maturities move by equal amounts
Slope	Rotational shift — short-term rates move more than long-term rates

Factor	Effect on Yield Curve
Curvature	Hump shift — medium-term rates change most

Macroeconomic Drivers

- **Monetary policy (Fed funds rate)** → controls the short end; explains 80–90% of slope factor movement
- **Inflation expectations** → drives the level factor; 5-year avg. core CPI explains ~66% of level variability
- **Real economic activity** → affects medium-term yields directly; long-term effects via level factor
- Contractionary monetary policy → slope ↓ (curve flattens/inverts) because short rates rise but long rates embed falling future rate expectations

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5. What Caused the Decline in Long-Term Yields? (FRBSF, 2013)

Two Components of Long-Term Interest Rates

$$\text{Long-term yield} = \underbrace{\text{Expectations component}}_{\text{Future short-term rates}} + \underbrace{\text{Risk premium (term premium)}}_{\text{Compensation for uncertainty}}$$

Key Findings (Bauer, Rudebusch & Wu, 2013 — Bias-Corrected Method)

Component	Standard Model (biased)	Bias-Corrected Model
Expectations	Stable — no trend	Pronounced downward trend (~3 pp decline)
Risk Premium	Massive decline (5% → −1%)	Smaller decline (~3 pp); **countercyclical**

Main Conclusion

- The secular decline in 10-year yields (8% → 2%, 1990–2012) was driven primarily by:
    1. **Falling long-run inflation expectations** (~2 pp contribution)
    2. **Lower expected real interest rates** (~1 pp contribution)
    3. Some reduction in the risk premium (less than standard models suggest)
- Standard models **overestimate** the role of risk premium reduction due to statistical bias (rates revert to mean too fast)

- The bias-corrected risk premium is **countercyclical** — rises in recessions, falls in expansions

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6. Recession Prediction Using the Yield Curve (FRBSF, 2022)

The Yield Curve as Recession Indicator

- The yield curve has inverted **before all 10 U.S. recessions since 1955**
- Inversion typically precedes recession by **12–18 months**

Two Competing Measures (as of 2022)

Measure	Behavior (2022)	Recession Signal
10y − 2y spread	Approaching zero / inverted briefly	Elevated risk (~17% probability)
10y − 3m spread	Widening (positive)	Low risk (~4% probability)

Which Spread Is Better?

Academic research and empirical evidence favor the 10y − 3m spread because: 1. It captures **full near-term rate expectations** (including expected Fed hikes in coming 2 years) 2. The 10y − 2y spread "differences out" near-term rate expectations — missing current Fed tightening signals 3. Statistically, 10y − 3m **outperforms** 10y − 2y in recession prediction over 4+ decades of data 4. An alternative: **Near-term forward spread** (18–21 month forward rate minus 3m rate) — focuses on near-future policy rate; also has strong predictive power

Critical Thresholds

Spread	Zero-spread threshold (recession signal)	Implied recession prob. at threshold
10y − 2y	0% spread	~22% probability
10y − 3m	0% spread	~32% probability

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7. Uses of the Yield Curve

1. **Leading economic indicator** — predicts recessions and expansions

2. **Benchmark for pricing** — all fixed-income securities priced as spread "over the curve"
    - e.g., high-quality corporate bond = Treasury yield + 60 bps
3. **Portfolio strategy** — managers anticipate curve moves to earn above-average bond returns

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8. Empirical Project Guidelines (for Reference)

- Choose **one publicly listed company** (unique — not duplicated)
- Provide brief company profile and motivation
- Attach **3 years of financial statements** (2022–2024)
- Calculate and interpret **financial ratios**
- Draw conclusions: company status + policy implications
- Submission: Typed, submitted to professor's office **before last class day**

Key Formulas to Remember

Formula	Expression
Current Yield	Coupon / Purchase Price
Term Spread	Long-term yield − Short-term yield
Long-term Yield	Expectations Component + Risk Premium
Preferred recession predictor	**10y − 3m Treasury spread**

Exam Focus Areas (High Priority)

1. **Four yield curve shapes** → know the shape, cause, and economic implication of each
2. **Three theories** explaining yield curve shape (Pure Expectations, Liquidity Preference, Preferred Habitat)
3. **Three factors** driving yield curve movements (Level, Slope, Curvature) and their macroeconomic links
4. **Two-component decomposition** of long-term yields: expectations vs. risk premium
5. **Bias-corrected vs. standard model** — why standard methods overstate risk premium's role
6. **10y—3m vs. 10y—2y spread** — which is preferred for recession prediction and why
7. **Countercyclical risk premium** — rises in recessions, falls in expansions